

Experiment 2: Candidate Elimination Algorithm

Aim

To implement the Candidate Elimination algorithm in Python to determine the set of all hypotheses that are consistent with the given training examples.

Algorithm

Step 1: Import the required libraries.

Step 2: Read the training dataset from the trainingdata.csv file.

Step 3: Initialize the Specific hypothesis (S) with the first training example.

Step 4: Initialize the General hypothesis (G) with the most general hypothesis.

Step 5: For each training example:

- If the example is positive ("Yes"):
 - Generalize the Specific hypothesis.
 - Remove inconsistent hypotheses from the General hypothesis.
- If the example is negative ("No"):
 - Specialize the General hypothesis to exclude the negative example.

Step 6: Remove duplicate and overly general hypotheses.

Step 7: Display the final Specific and General hypotheses.

Program

```
import csv

with open('trainingdata.csv') as csvFile:
    data = list(csv.reader(csvFile))

concepts = [x[:-1] for x in data[1:]]
target = [x[-1] for x in data[1:]]

specific = concepts[0]
general = ["?" for _ in range(len(specific)) for _ in range(len(specific))]

for i, h in enumerate(concepts):

    if target[i] == "Yes":

        for x in range(len(specific)):
            if h[x] != specific[x]:
                specific[x] = '?'
                general[x][x] = '?'

    elif target[i] == "No":

        for x in range(len(specific)):
            if h[x] != specific[x]:
                general[x][x] = specific[x]
            else:
                general[x][x] = '?'

general = [g for g in general if g != ['?'] * len(specific)]

print("Specific Hypothesis:")
print(specific)

print("\nGeneral Hypothesis:")
for g in general:
    print(g)
```

Output

```
... Specific Hypothesis:
['Sunny', 'Warm', '?', 'Strong', '?', '?']

General Hypothesis:
['Sunny', '?', '?', '?', '?', '?']
['?', 'Warm', '?', '?', '?', '?']
```

Result

Thus, the Candidate Elimination algorithm was successfully implemented, and the Specific and General hypotheses consistent with the given training data were obtained.